LINFO 1113 : Algorithmique Numérique

Initial Value Problems

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Outline for today

- Initial Value Problem
- Euler's Method
- Runge-Kutta Methods
  - 2nd order
  - 4th order
- Stability and Stiffness
- Systems of differential equations

Problem statement and terminology

Initial Value Problem (IVP, also named Cauchy problem)

First-order differential equation (ODE)

$$\begin{aligned} y'(x) &= f(x, y(x)), \qquad x \in [a, b] \\ y(a) &= \bar{y} \end{aligned}$$

where $y' = \dfrac{dy}{dx}$ and $f(x, y)$ is a given function, $a, b, \bar{y} \in \mathbb{R}$ are given.

Applications:

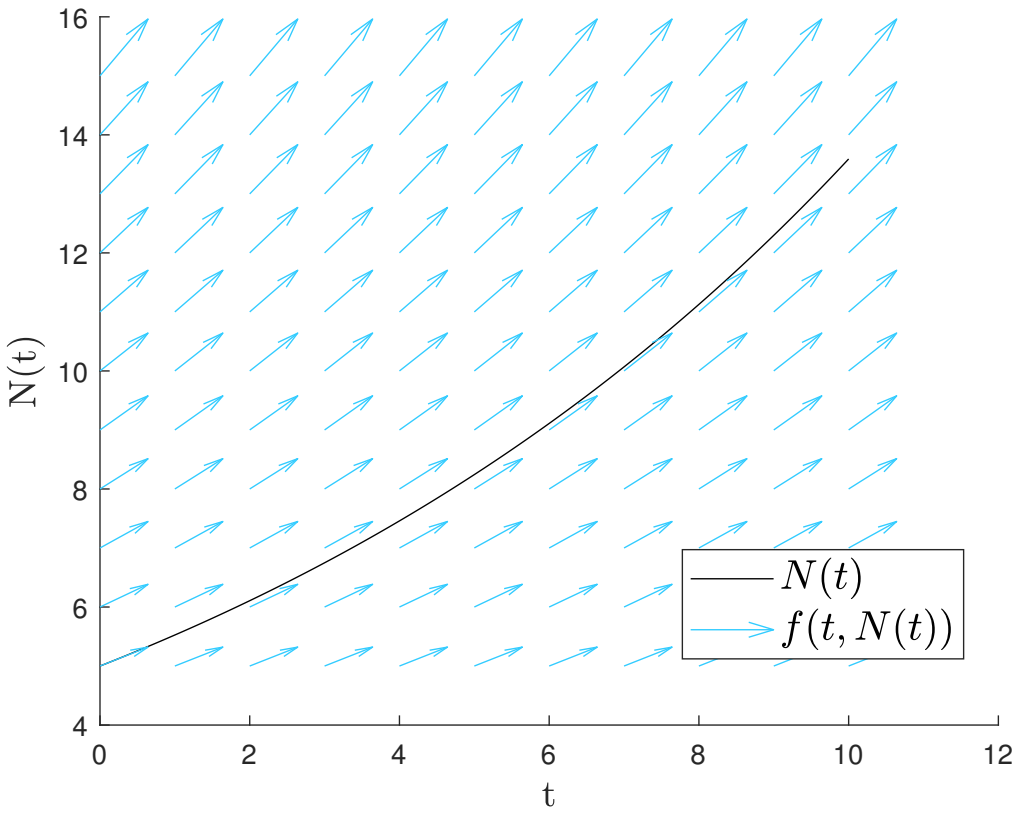
- Evolution of the capacitor charge in an electric circuit
- Population growth in ecological or biological systems
- Mechanical models: modeling the movement of a robotic arm
- Predatory-prey models in ecological systems

Example: population models

Model:

$$N'(t) = rN(t), \qquad t \in [0, T]$$
$$N(0) = \bar{N}$$

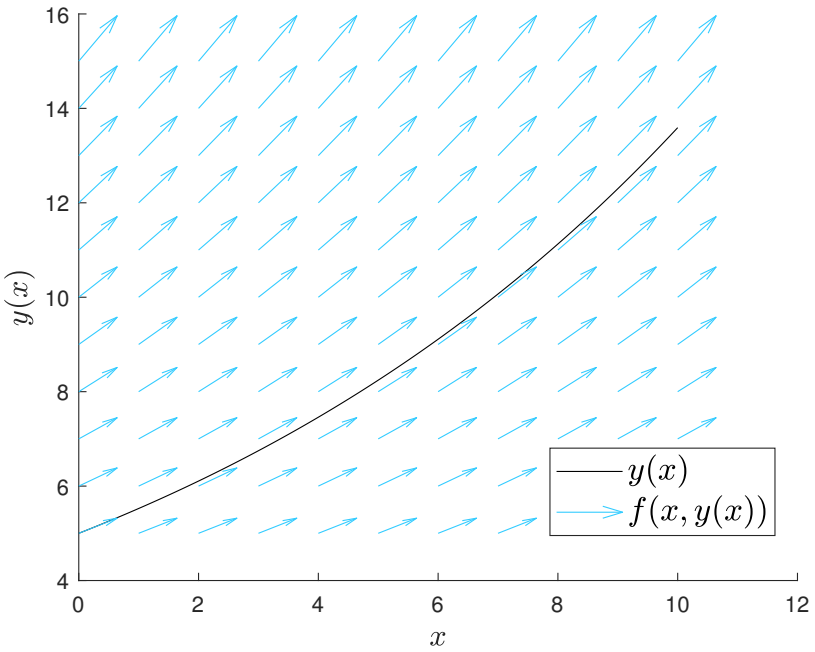
where r is the growth rate.

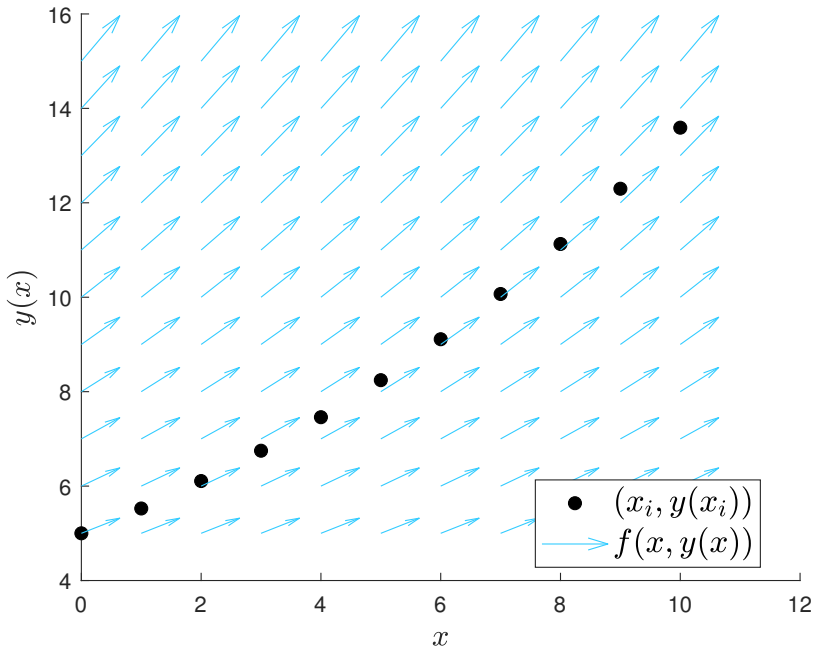


Approximating numerically the solution to an IVP

Instead of computing a continuous curve, we will aim to approximate its value at $x_1, \ldots, x_n$ such that

$$x_{i+1} - x_i = h \qquad \forall i = 0, \ldots, n-1.$$





The goal is then to find a sequence $y_1, \ldots, y_n$ that is as close as possible to the sequence $y(x_1), \ldots, y(x_n)$.

Euler's method

- Euler's method is conceptually simple
- It starts from the truncated Taylor development

Taylor's formula with remainder

Let y be an m-times continuously differentiable function on an interval containing x and z. Then

$$y(z) = y(x) + y'(x)(z-x) + \cdots + y^{(m-1)}(x)\frac{(z-x)^{m-1}}{(m-1)!} + y^{(m)}(\xi)\frac{(z-x)^m}{m!},$$

where ξ lies between x and z.

- Taking $m = 2$, $x = x_i$ and $z = x_{i+1}$ gives:

$$y(x_{i+1}) = y(x_i) + y'(x_i)(x_{i+1} - x_i) + y''(\xi)\frac{(x_{i+1} - x_i)^2}{2},$$

where ξ lies between x_i and x_{i+1}.

Euler's method

We approximate:

$$y(x_{i+1}) \simeq y(x_i) + y'(x_i)(x_{i+1} - x_i),$$

i.e., we neglect the second-order term $y''(\xi)\dfrac{(x_{i+1} - x_i)^2}{2}$.

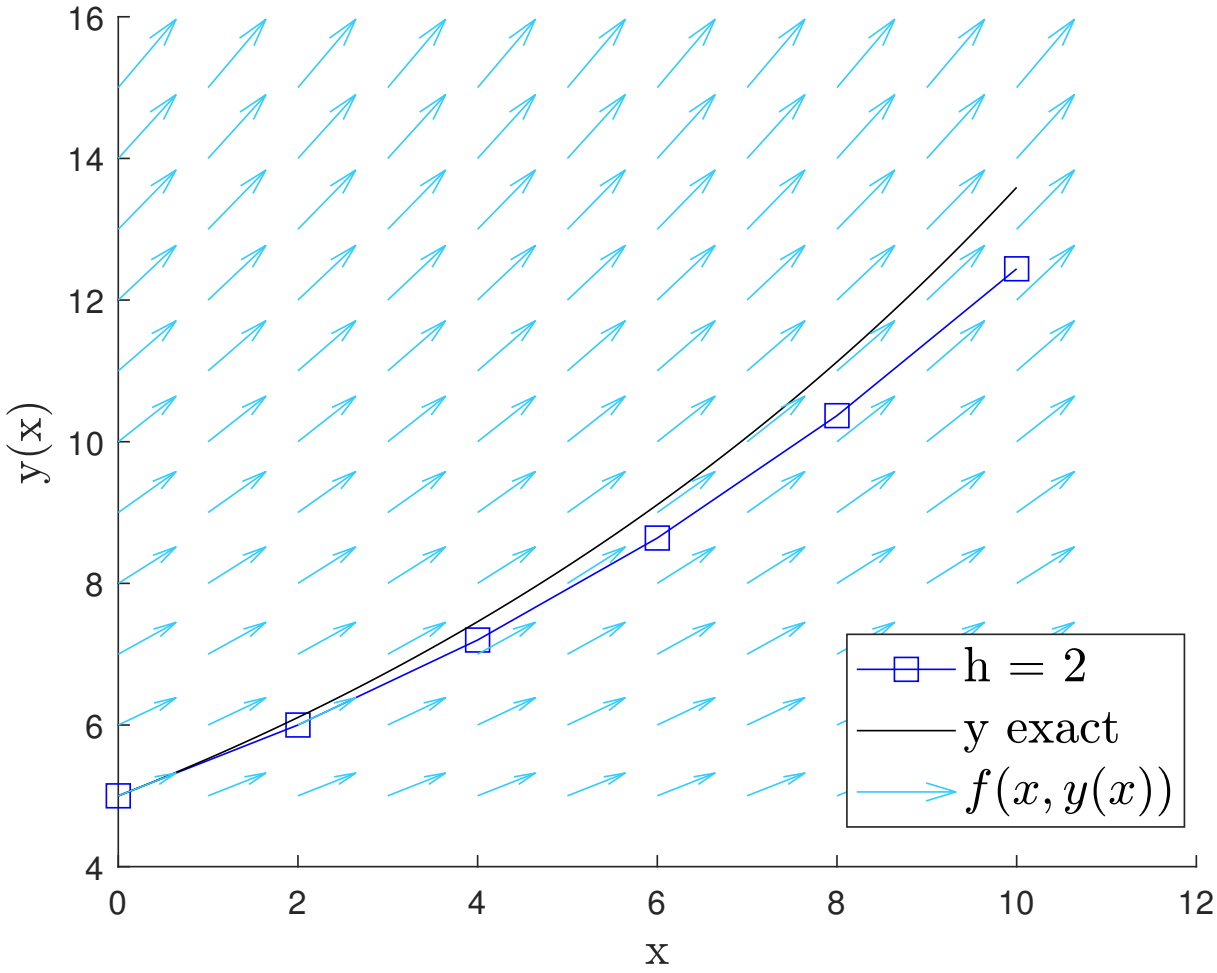
We replace $y(x_i)$ by y_i and $y'(x_i)$ by $f(x_i, y_i)$, to get:

Euler's method for solving a Cauchy problem

$$\begin{cases} y_0 &= y(x_0) \\ y_{i+1} &= y_i + hf(x_i, y_i). \end{cases}$$

Euler's method: example

$$\begin{cases} y'(x) &= 0.1y(x) \\ y(x_0) &= 5 \end{cases}$$



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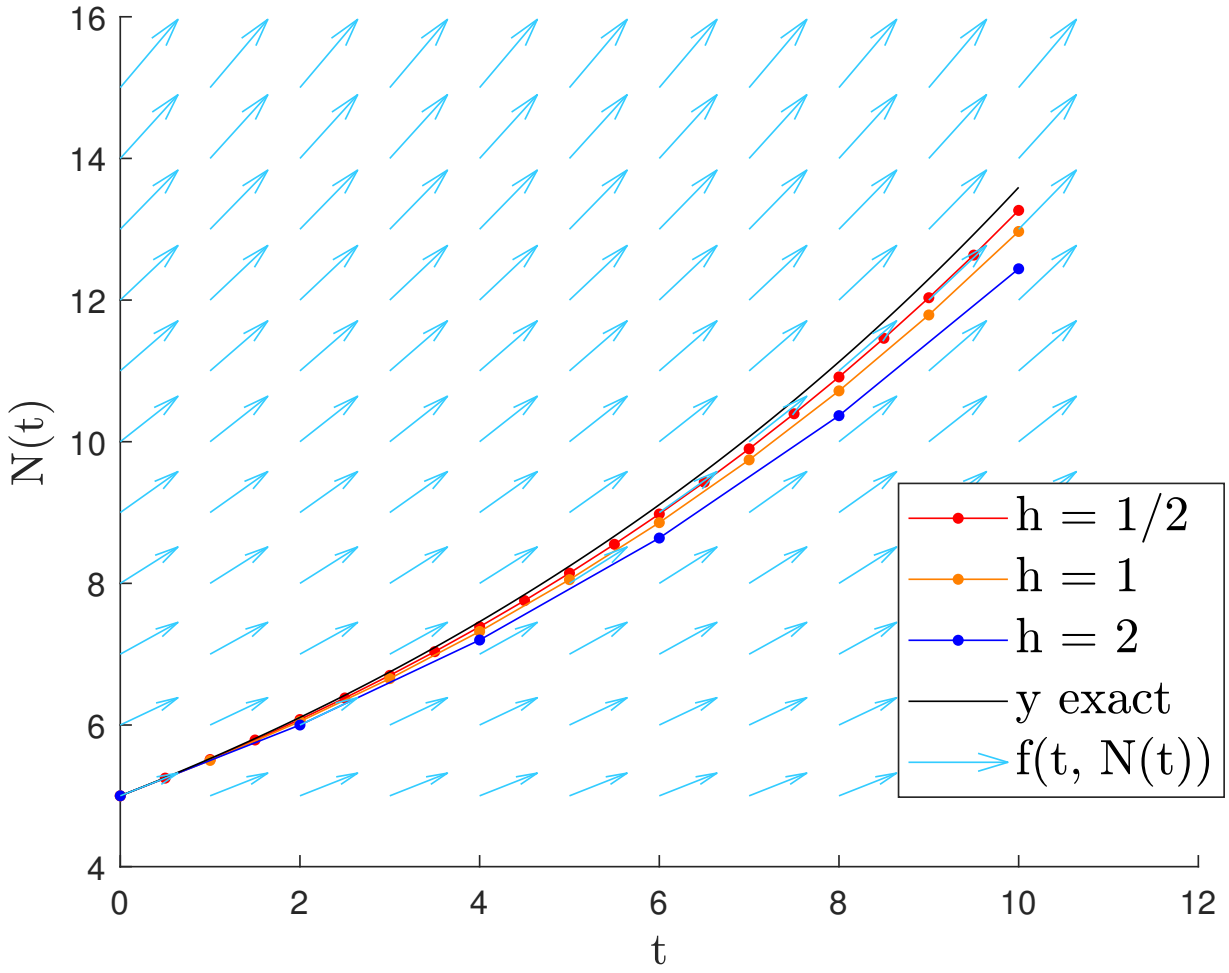


Approximation error for Euler's method

From the Taylor development, we get:

$$\underbrace{y(x_{i+1}) - y_{i+1}}_{E_{i+1}} = \underbrace{y(x_i) - y_i}_{E_i} + h\underbrace{(y'(x_i) - f(x_i, y_i))}_{\propto E_i} + \underbrace{h^2 \frac{y''(\xi_i)}{2}}_{E_i^{Tr}},$$

where ξ_i lies between x_i and x_{i+1}. The truncation error in Taylor's development is

$$E_i^{Tr} = \mathcal{O}(h^2) \qquad \forall i.$$

Error propagation from one point to another further increases the error, so that the total error can be shown to be (under suitable regularity assumptions on f):

$$E_i = \mathcal{O}(h) \qquad \forall i$$

(we say that Euler's method is **first-order accurate**).

Example

- Solve the initial value problem given by:

$$\begin{cases} y' + 4y &= x^2 \\ y(0) &= 1 \end{cases},$$

  by using Euler's method with $h = 0.01$, from $x = 0$ to 0.02.
- Compare your findings to the true solution and compute the accumulated error at each step. For this, use the analytical solution:

$$y = \frac{31}{32}e^{-4x} + \frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{32}.$$

Runge-Kutta method - 2nd order

Runge–Kutta methods aim to achieve higher accuracy by sacrificing the efficiency of Euler's method through re-evaluating f at points intermediate between (x_i, y_i) and (x_{i+1}, y_{i+1}).

2nd-order Runge-Kutta method

$$y_{i+1} = y_i + h(aK_0 + bK_1),$$

where

$$\begin{aligned} K_0 &= f(x_i, y_i) \\ K_1 &= f(x_i + \alpha h, y_i + \beta h K_0), \end{aligned}$$

and where the parameters a, b, α and β are to be determined.

Note that Euler's method can be seen as a member of this family of methods, with $a = 1$ and $b = 0$.

Runge-Kutta method - 2nd order

- The parameters are obtained by identifying the resulting estimation of $y(x_{i+1})$ with a second-order truncated Taylor development of y (i.e., with error term in $\mathcal{O}(h^3)$).
- This does not lead to a unique choice... Three choices are common for 2nd order Runge-Kutta methods:

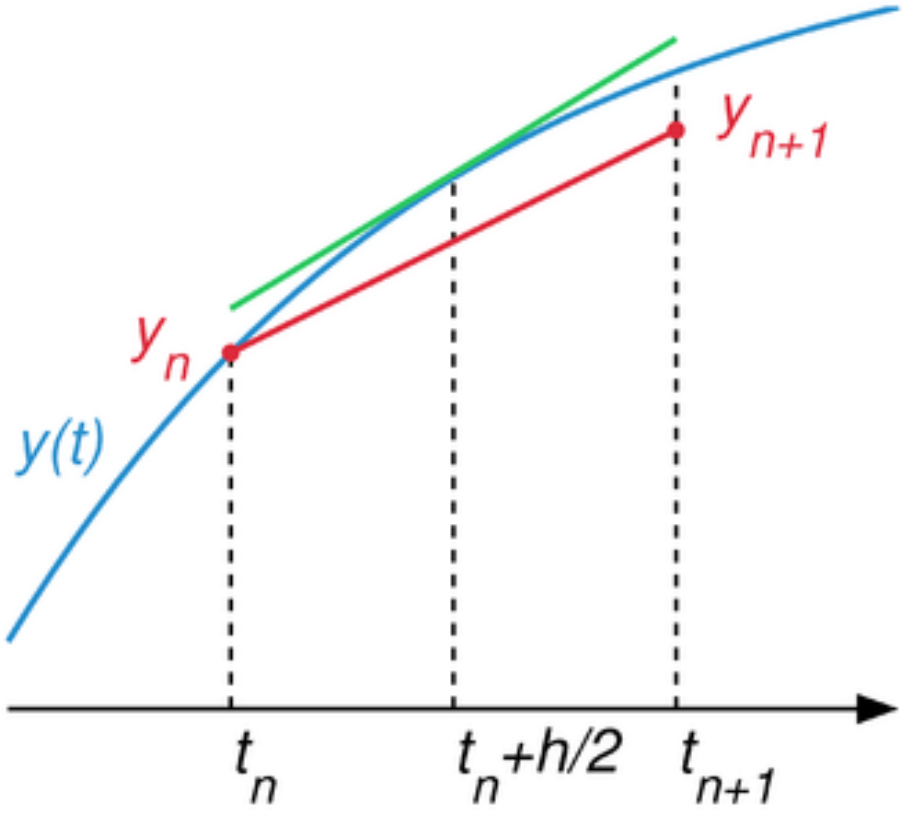
- $a = 0$, $b = 1$, $\alpha = 1/2$, $\beta = 1/2$: Modified Euler's method
- $a = 1/2$, $b = 1/2$, $\alpha = 1$, $\beta = 1$: Heun's method
- $a = 1/3$, $b = 2/3$, $\alpha = 3/4$, $\beta = 3/4$: Ralston's method

- All these methods have a truncation error in $\mathcal{O}(h^3)$
- They all have a global error of $\mathcal{O}(h^2)$ (i.e., they are second-order methods).

Modified Euler's method

$$y_{i+1} = y_i + hf(x_i + \frac{h}{2}, y_i + \frac{h}{2}f(x_i, y_i))$$

- Also called midpoint method.
- Take the tangent of f at midpoint to propagate y.



Example

- Solve the initial value problem given by:

$$\begin{cases} y' + 4y &= x^2 \\ y(0) &= 1 \end{cases},$$

  by using the modified Euler's method with $h = 0.01$, from $x = 0$ to 0.01.

- Compare your findings to the true solution and compute the accumulated error at each step. For this, use the analytical solution:

$$y = \frac{31}{32}e^{-4x} + \frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{32}.$$

Connection to integration methods

- In the special case where $f(x, y)$ only depends on x, the Cauchy problem can be written as

$$\begin{cases} y'(x) &= f(x), \qquad x \in [a, b] \\ y(a) &= \bar{y} \end{cases}.$$

- Mathematically, $y(x_{i+1})$ is given by

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(x)dx.$$

- If we compare with the Modified Euler's method, we get:

$$y(x_{i+1}) \simeq y(x_i) + hf\left(x_i + \frac{h}{2}\right)$$

(i.e., we use the midpoint integral estimation)

- If we compare with Heun's method, we get:

$$y(x_{i+1}) \simeq y(x_i) + \frac{h}{2}(f(x_i) + f(x_i + h))$$

(i.e., we use the trapezoid integral estimation)

Runge-Kutta method - 4th order

Runge-Kutta of 4th order (RK4)

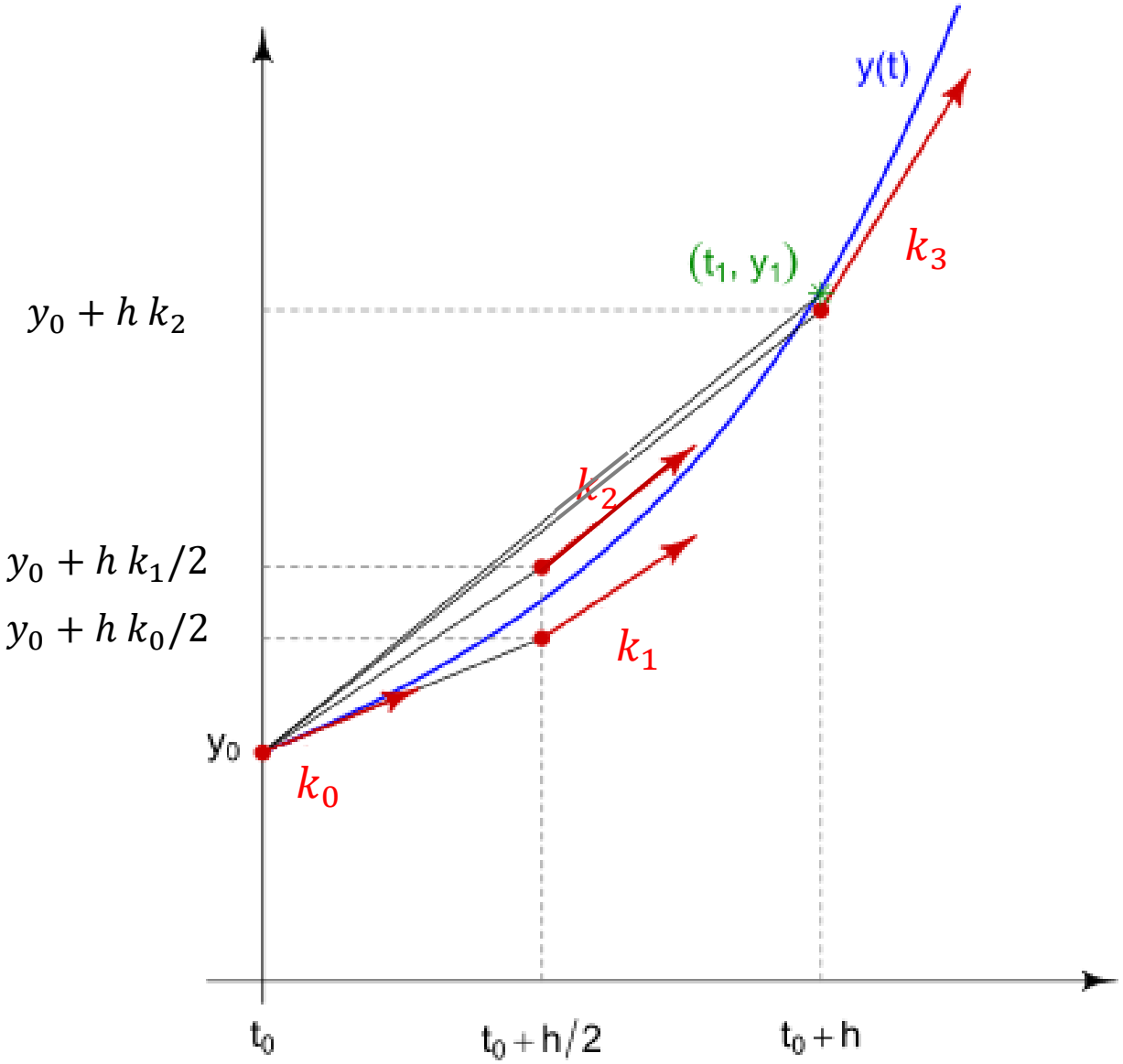
$$y_{i+1} = y_i + \frac{h}{6}(K_0 + 2K_1 + 2K_2 + K_3).$$

with

$$\begin{cases} K_0 &= f(x_i, y_i) \\ K_1 &= f\left(x_i + \frac{h}{2}, y_i + h\frac{K_0}{2}\right) \\ K_2 &= f\left(x_i + \frac{h}{2}, y_i + h\frac{K_1}{2}\right) \\ K_3 &= f(x_i + h, y_i + hK_2) \end{cases},$$

- The coefficients were obtained by matching a 4th-order Taylor development of y.
- The local truncation error is $\mathcal{O}(h^5)$.
- The global error of the method is $\mathcal{O}(h^4)$.

Runge-Kutta method - 4th order



Example

- Solve the initial value problem given by:

$$\begin{cases} y' + 4y &= x^2 \\ y(0) &= 1 \end{cases},$$

  by using RK4 with $h = 0.01$, from $x = 0$ to 0.01.
- Compare your findings to the true solution and compute the accumulated error at each step. For this, use the analytical solution:

$$y = \frac{31}{32}e^{-4x} + \frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{32}.$$

Stability

- A numerical method is stable if
  - The effects of local errors do not accumulate catastrophically;
  - The global error remains bounded.
- If the method is unstable, the global error will increase exponentially, eventually causing numerical overflow.
- Stability has nothing to do with accuracy;
- An inaccurate method can be very stable.
- Stability is determined by three factors:
  - the differential equations,
  - the method of solution,
  - the value of the increment h

Stability of Euler's method

Consider the Cauchy problem:

$$\begin{cases} y'(x) &= -\lambda y(x) \\ y(0) &= \beta \end{cases},$$

with $\lambda > 0$. The analytical solution is given by:

$$y(x) = \beta e^{-\lambda x}.$$

Let us solve the system via Euler's method:

$$y_{i+1} = y_i + hf(x_i, y_i)$$

Using $f(x, y) = -\lambda y$, we get:

$$y_{i+1} = y_i(1 - \lambda h).$$

If $|1 - \lambda h| > 1$, the method is clearly unstable, because y increases at every iteration.

Euler's method is stable only if $h \leq 2/\lambda$.

ODEs of order n

An ordinary differential equation (ODE) of order n

$$y^{(n)} = f(x, y, y', y'', \ldots, y^{(n-1)})$$

can always be transformed into n first-order ODEs.

Let us write:

$$\begin{aligned} y[0] &= y \\ y[1] &= y' \\ y[2] &= y'' \\ &\vdots \\ y[n-1] &= y^{(n-1)}. \end{aligned}$$

Remark : ODEs of order n

The ODE of order n can be written as:

$$\begin{aligned} y[0]' &= y[1] \\ y[1]' &= y[2] \\ &\vdots \\ y[n-1]' &= f(x, y[0], y[1], \ldots, y[n-1]). \end{aligned}$$

The solution requires n-initial values:

$$\begin{aligned} y[0](a) &= \alpha[0], \\ y[1](a) &= \alpha[1], \\ &\vdots \\ y[n-1](a) &= \alpha[n-1]. \end{aligned}$$

Remark : ODEs of order n

These equations can be rewritten in a concise way, using matrix notation:

$$\begin{cases} \mathbf{y}' = \mathbf{F}(x, \mathbf{y}) \\ \mathbf{y}(a) = \boldsymbol{\alpha} \end{cases}$$

where

$$F(x, \mathbf{y}) = \begin{pmatrix} y[1] \\ y[2] \\ \vdots \\ f(x, y[0], y[1], \ldots, y[n-1]) \end{pmatrix}$$

Example - ODEs of order n

Apply Euler's method to the second-order differential equation

$$y'' + y' - 5y = 0,$$

on the interval $[0, 1]$ with initial conditions $y(0) = 0, y'(0) = 1$.

Remark: Boundary value problems

In some cases, y is specified at different values of x

- This is called a boundary value problem
- This is more complex than an IVP
- Not discussed in this course

Examples:

- $y'' = -y, \qquad y(0) = 1, \qquad y'(0) = 0$ is an initial value problem
- $y'' = -y, \qquad y(0) = 1, \qquad y(\pi) = 0$ is a boundary value problem

Stiffness

- A Cauchy problem is stiff if some terms in the solution vector $\mathbf{y}(x)$ grow much more rapidly with x than others.
- Stiffness can be easily predicted for the differential equations $\mathbf{y}' = -\mathbf{A}\,\mathbf{y}$ with constant coefficient matrix $\mathbf{A}$.
- The solution of these equations is $\mathbf{y}(x) = \sum_i C_i \mathbf{v}_i e^{-\lambda_i x}$,

  where λ_i are the eigenvalues of $\mathbf{A}$ and $\mathbf{v}_i$ are the corresponding eigenvectors.
- It is evident that the problem is stiff if there is a large disparity in the magnitudes of the positive eigenvalues.
- Numerical integration of stiff equations requires special care.
- The step size h needed for stability is determined by the largest eigenvalue $\lambda_{\max}$, even if the terms $e^{-\lambda_{\max} x}$ in the solution decay very rapidly and become insignificant as we move away from the origin.